AI ACADEMY · LEVEL 1 · GRADE 6 · WEEK 7 · WORKBOOK

Categories and Labels

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

THE CONNECTED WEEK

One connected label-and-policy record.

Apply a guide, clarify a vague instruction, inspect annotator disagreement and retain the policy version.

Keep one habit: Keep the first labels, reasons and corrections distinguishable.

MAIN LESSON · SECTION 1

Recall: reconnect earlier ideas

> Retrieve before opening notes.

Retrieve and explain

- Try the recall before opening notes. Then use a different colour or a **label**led second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Sound Becomes a Signal. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Rules or Learned Patterns?. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from AI or Not?. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

My remembered idea, example and correction

My remembered idea, example and correction

▼ Earlier wording, alternatives and picture transcripts

Sound Becomes a Signal

Rules or Learned Patterns?

AI or Not?

W1 · Apply the calendar guide

> Cite the deciding field and criterion for every card.

Your evidence record

- The calendar policy accepts whole-number days 1 through 30 inclusive and a room code A or B. Check known failures first, then unknown fields, then readiness.

  Label these cards and give **evidence**: day 30 room B; day 31 room A; day 14 room unreadable; day 31 room unreadable.

> **My reasoning and evidence**

MAIN LESSON · SECTION 13

W2 · Replace a vague instruction

> Write a reproducible rule including blank fields.

Your evidence record

- A helper says **Label** a card ready if it looks nice. Explain why this can lead to inconsistent **label**s.

  Replace it with an instruction that another person can apply using the calendar policy, including what to do about a blank field.

My reasoning and evidence

MAIN LESSON · SECTION 14

W3 · Inspect agreement and disagreement

> Count matching positions before judging their correctness.

Your evidence record

- Use W1's card order: day 30 room B; day 31 room A; day 14 room unreadable; day 31 room unreadable. Compare the two supplied annotator lists position by position. Use the calendar policy separately when checking which labels follow it.

- For the same four cards, Mina records ready, revise, review, ready. Theo records ready, review, review, revise. How many positions agree? Identify the **disagreement**s.

  Can you decide which **annotator** is correct from these **label** lists alone? Explain what else you need.

My reasoning and evidence

MAIN LESSON · SECTION 16

W5 · Keep the policy version

> Trace what changes when the accepted range changes.

Your evidence record

- Version 1 accepts calendar days 1 through 30. Version 2 accepts days 1 through 31. A card says day 31 room A. State its **label** under each version.

  Explain why mixing the two **label**s without recording the version could create a misleading **training** dataset.

> **My reasoning and evidence**

MAIN LESSON · SECTION 4

W4 · Transfer the method to a new record

> Keep one first attempt before hints or feedback.

Read and explain independently

A fictional equipment-request form accepts a whole-number borrowing duration of 1 through 7 days inclusive and one item code X or Y.

Label revise if any known field fails; otherwise **label** review if a field is blank or unreadable; otherwise **label** ready. Classify these forms and justify each: 7 days X; 8 days Y; unreadable duration X; 8 days blank code.

Explain whether ready alone grants permission to borrow.

> **My independent explanation, trace and limits**
>
> _____
>
> _____
>
> _____
>
> _____

Keep a fair record

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Transfer the method to a new record

MAIN LESSON · SECTION 9

Project · Keep the decision and its evidence

> Add this week's change to your existing project record.

Your evidence record

Define the Classroom Sorter's bin **label**s and write how to handle objects that reasonably fit more than one bin.

The project change and **evidence** I will keep

Write or draw your reasoning and evidence.

MAIN LESSON · SECTION 17

Feedback · Repair and return

> Keep the first attempt, the repair and a changed-case return.

Your evidence record

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

MAIN LESSON · SECTION 10

Finish · Judge the evidence in your work

> Point to work that supports your confidence level.

Your evidence record

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> This is the same W2 attempt. Keep one response record.

Use this when helpful

Use W2: Replace a vague instruction for Categories and operational **label**s. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

▼ Extra practice · Repair the explanation

REFERENCE · SECTION 3

Extra practice · Repair the explanation

> Explain the deciding fact before discussing the answer.

Use this when helpful

Use the worked case for Categories and operational **label**s. Proposed explanation: “A blank is the same as a failing value”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Original learning target

REFERENCE · SECTION 5

Reference · Original learning target

> Use this target to inspect an explanation.

Use this when helpful

Your dataset **label**s 'reusable containers'. One team includes a disposable bottle reused once; another excludes it. Give an operational **label** rule and classify this boundary case under that rule.

Explain why a different rule could be defensible.

▼ **Extra practice · A complete case**

REFERENCE · SECTION 6

Extra practice · A complete case

> Try the question before checking the worked explanation.

Use this when helpful

Six cards show two circles, two triangles and two squares. Sort by shape.

Why can red and blue circles share a **label**?

Show your trace, calculation or **evidence**

> **Write or draw your reasoning and evidence.**

▼ Extra practice · A changed case

REFERENCE · SECTION 7

Extra practice · A changed case

> Keep this practice separate from the reserved independent attempt.

Use this when helpful

For this toy sorter, label an object Seed-bearing when seeds are observed, Other when inspection establishes that it has no seeds, and Review when the needed contents cannot be observed. Sort an opened apple with visible seeds, a pencil and a sealed mystery box. Which item needs more **evidence**, and why?

My practice answer and reason

Write or draw your reasoning and evidence.

Week 7 Test and reflect

▼ Extra practice · Design a revealing test

REFERENCE · SECTION 8

Extra practice · Design a revealing test

> Predict before tracing or testing.

Use this when helpful

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

▼ **Reference · Original target response space**

REFERENCE · SECTION 11

Reference · Original target response space

> This teaching case is practice rather than a fresh assessment.

Use this when helpful

Your dataset **label**s 'reusable containers'. One team includes a disposable bottle reused once; another excludes it. Give an operational **label** rule and classify this boundary case under that rule.

Explain why a different rule could be defensible.

My explanation after practice

Record support used: none, hint or teacher explanation.

▼ **Reference · Independent-case response copy**

REFERENCE · SECTION 15

Reference · Independent-case response copy

> This duplicates W4. Reuse the same first-attempt record.

Use this when helpful

A fictional equipment-request form accepts a whole-number borrowing duration of 1 through 7 days inclusive and one item code X or Y.

Label revise if any known field fails; otherwise **label** review if a field is blank or unreadable; otherwise **label** ready. Classify these forms and justify each: 7 days X; 8 days Y; unreadable duration X; 8 days blank code.

Explain whether ready alone grants permission to borrow.

My reasoning and evidence

▼ Extra practice · Use the worked picture

REFERENCE · SECTION 18

Extra practice · Use the worked picture

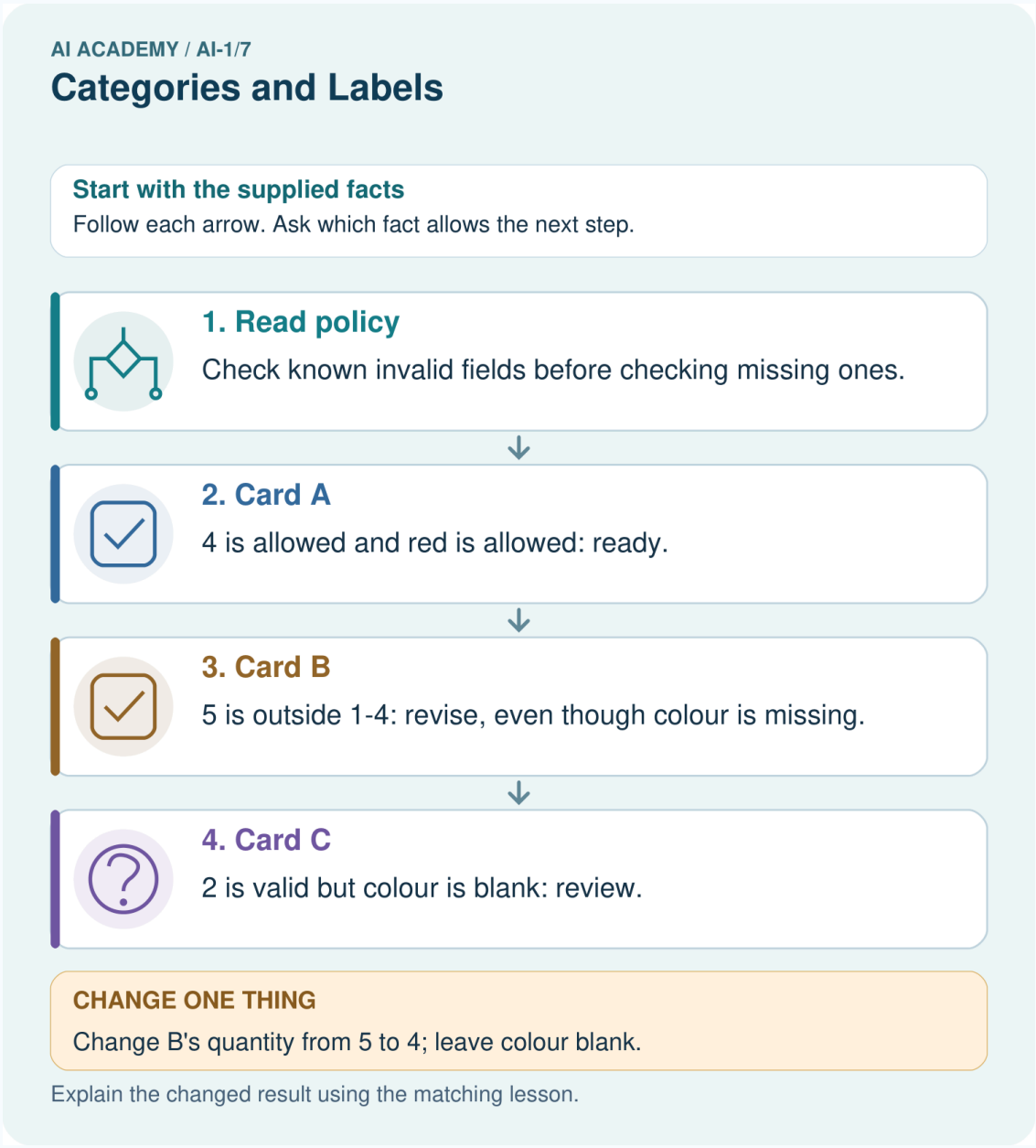
> Keep the explanation and the support you used.

Use this when helpful

A fictional craft club **label**s cardboard-order cards ready, review or revise.

Allowed quantity: whole number 1–4. Required colour: red or blue.

If any known field fails, revise; otherwise if a required field is blank, review; otherwise ready.
Cards A=(4,red), B=(5,blank), C=(2,blank).



Follow the labelled picture with the matching explanation. Open the vector diagram

Explain why B originally was revise rather than review, then **label** its changed version.

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: Write a sorter **label** guide with examples of complete, invalid and unresolved **input**s before comparing **label**s.

▼ **Reference · Project portfolio record**

REFERENCE · SECTION 19

Reference · Project portfolio record

> Extend your main project response when helpful.

Use this when helpful

Project: Helpful Classroom Sorter

Current milestone: Make a transparent first version. Keep a trace of how one **input** becomes an **output**; explain the rule or learned component.

Write a sorter **label** guide with examples of complete, invalid and unresolved **input**s before comparing **label**s.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ **Extra practice: pictures, repair and feedback**

REFERENCE · SECTION 20

Extra practice: pictures, repair and feedback

> Attempt each question before opening its explanation. Use one browser or paper copy.

First supported question

- A fictional craft club **label**s cardboard-order cards ready, review or revise.
- Which **label**led step matches this detail? 2 is valid but colour is blank: review.
- 1. Read policy
- 2. Card B
- 3. Card A
- 4. Card C
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Card C 2 is valid but colour is blank: review. Operational **label**s specify what a word means for this task.

Two **label**lers using the same priority should agree on these cards. Review describes missing **evidence**, not a hidden assumption that the card is wrong. Revise is supported by a known failure.

Changed-case question

- Changed-case practice: Change B's quantity from 5 to 4; leave colour blank. Which conclusion is supported by the supplied facts?
- 1. B becomes review.
- 2. We can decide the result without examining the changed **input** or the procedure.
- 3. This one example guarantees correct results for every future **input**.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the changed case**

After your attempt, compare with the supported explanation: B becomes review. The known failure is removed, but a required field is still missing.

Repair and save your work

Review Sound Becomes a Signal (ai-1/6). Goal: Interpret an ordered sequence of sound-signal samples with a stated time interval. Short practice: Does the all-zero retained record prove the full signal was silent?

Use the picture.. Return to this week and explain what you changed.

Review Rules or Learned Patterns? (ai-1/4). Goal: Execute a stated rule including its boundary. Short practice: Show which stage used **label**s and which stage used only the new measurement..

Return to this week and explain what you changed.

Review AI or Not? (ai-1/1). Goal: Distinguish an observed result from an explanation of how it was produced. Short practice: Point to the learned stage. Does the changed display prove the weather forecast changed?.

Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

▼ **Reference · Six-picture response sheets**

REFERENCE · SECTION 21

Reference · Six-picture response sheets

> Choose the image that helps rather than repeating all six.

01. Categories and Labels: the case

AI ACADEMY / AI-1/7 · 01 · LOOK AT THE FACTS

Categories and Labels: the case

The situation

A fictional craft club labels cardboard-order cards ready, review or revise.

Supplied evidence

Allowed quantity: whole number 1–4.
Required colour: red or blue.
If any known field fails, revise; otherwise if a required field is blank, review; otherwise ready. Cards A=(4,red), B=(5,blank), C=(2,blank).

By the end, I can apply a shared labelling policy consistently, including missing information.

Follow the labelled picture with the matching explanation. Open the vector diagram

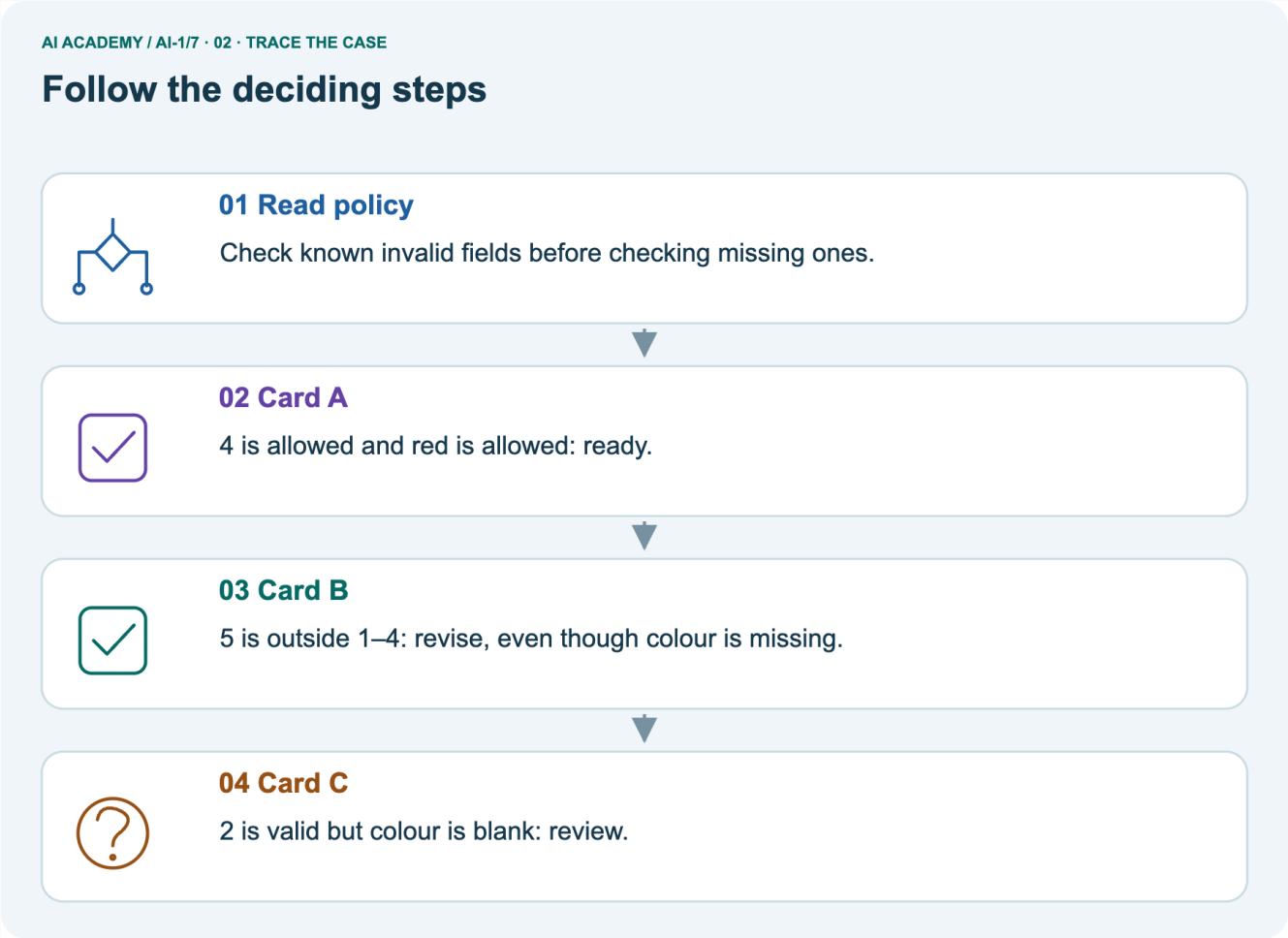
A fictional craft club **label**s cardboard-order cards ready, review or revise. Supplied facts: Allowed quantity: whole number 1–4. Required colour: red or blue.

If any known field fails, revise; otherwise if a required field is blank, review; otherwise ready. Cards A=(4,red), B=(5,blank), C=(2,blank).

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Read policy: Check known invalid fields before checking missing ones. → Card A: 4 is allowed and red is allowed: ready. → Card B: 5 is outside 1–4: revise, even though colour is missing. → Card C: 2 is valid but colour is blank: review.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Three ordered checks for a calendar card

AI ACADEMY / AI-1/7 · 03 · INSPECT THE LESSON MODEL

Three ordered checks for a calendar card

Check in this order	Evidence	Recorded label
First check known values	Any known field fails its criterion	Revise even if another field is unknown
Then check missing information	No known failure but a field is blank or unreadable	Review because evidence is incomplete
Otherwise all checks have passed	Day is 1 through 30 and room is A or B	Ready under this record policy

Stop at the first matching row. The order resolves a card with both a known failure and an unknown field. Ready describes the printed record under this policy; it does not confirm that the event exists or has permission to run.

Follow the labelled picture with the matching explanation. Open the vector diagram

Three ordered checks for a calendar card. First check known values | Any known field fails its criterion | Revise even if another field is unknown.

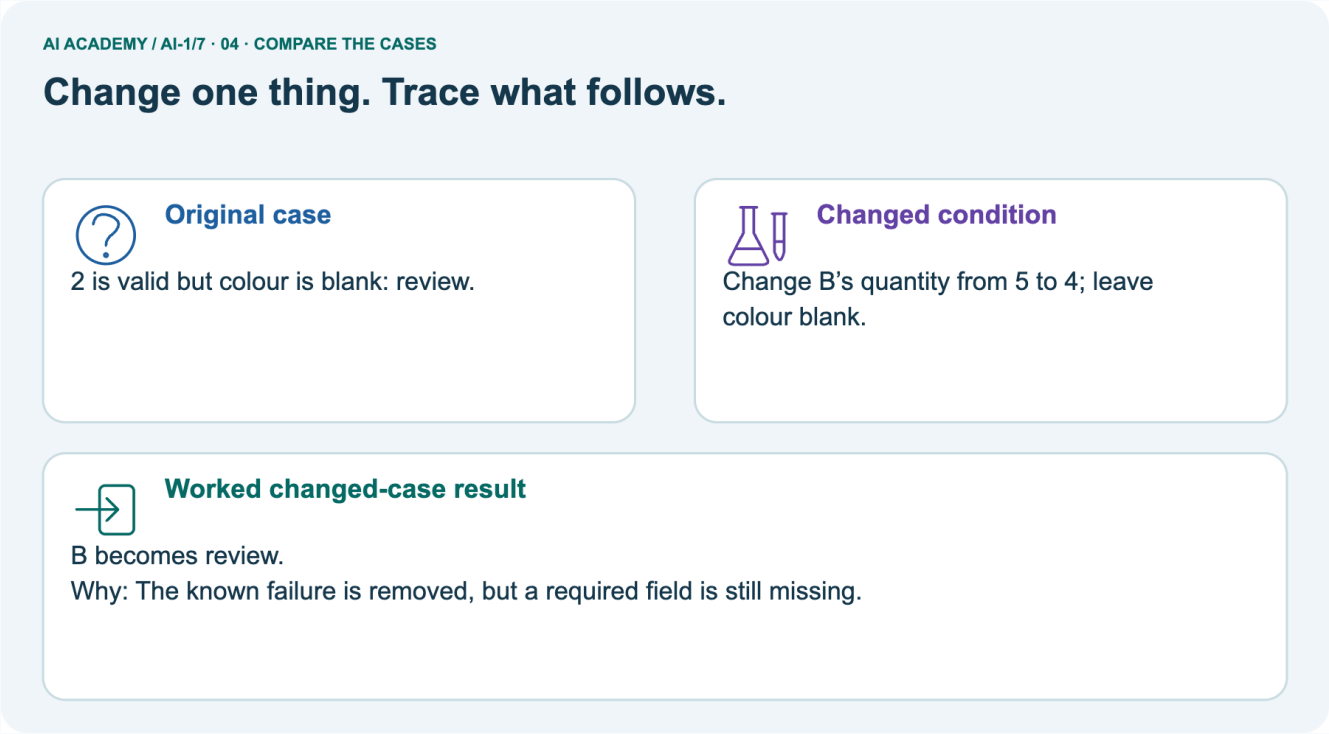
Then check missing information | No known failure but a field is blank or unreadable | Review because **evidence** is incomplete.

Otherwise all checks have passed | Day is 1 through 30 and room is A or B | Ready under this record policy

Read the **label**s and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

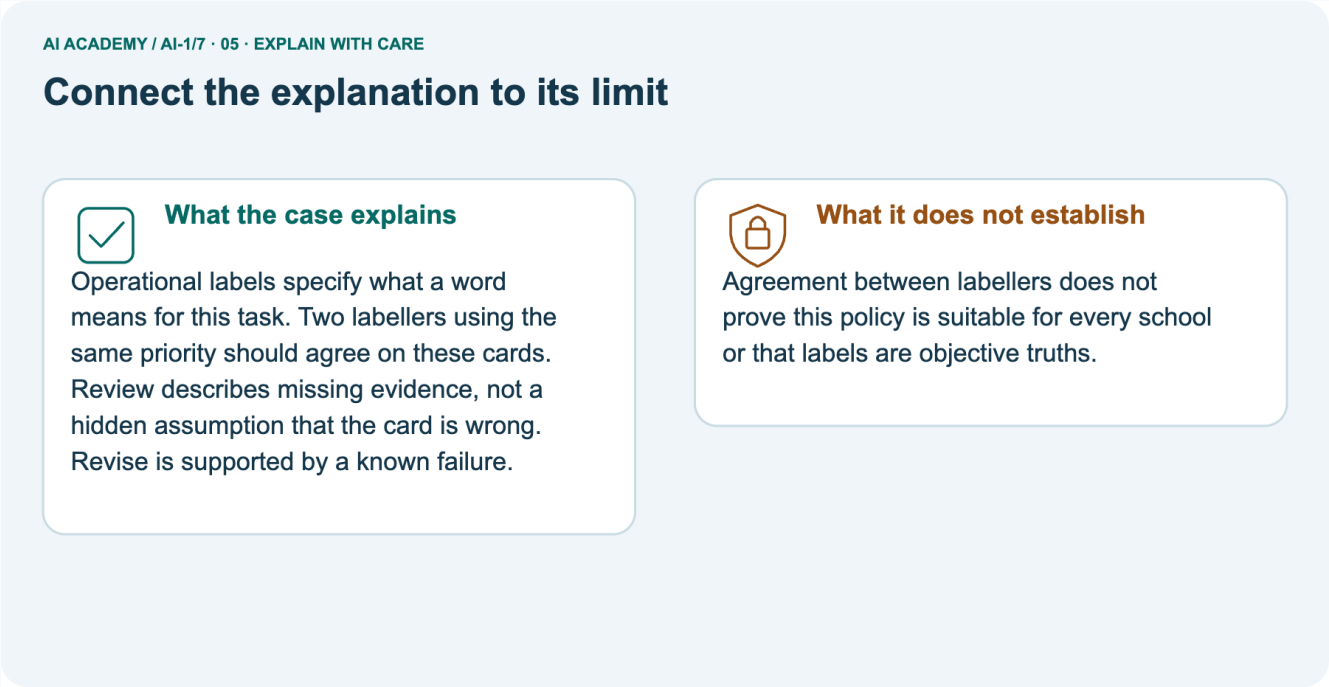
Original case: 2 is valid but colour is blank: review. Changed condition: Change B's quantity from 5 to 4; leave colour blank. Expected result: B becomes review.

Reason: The known failure is removed, but a required field is still missing.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

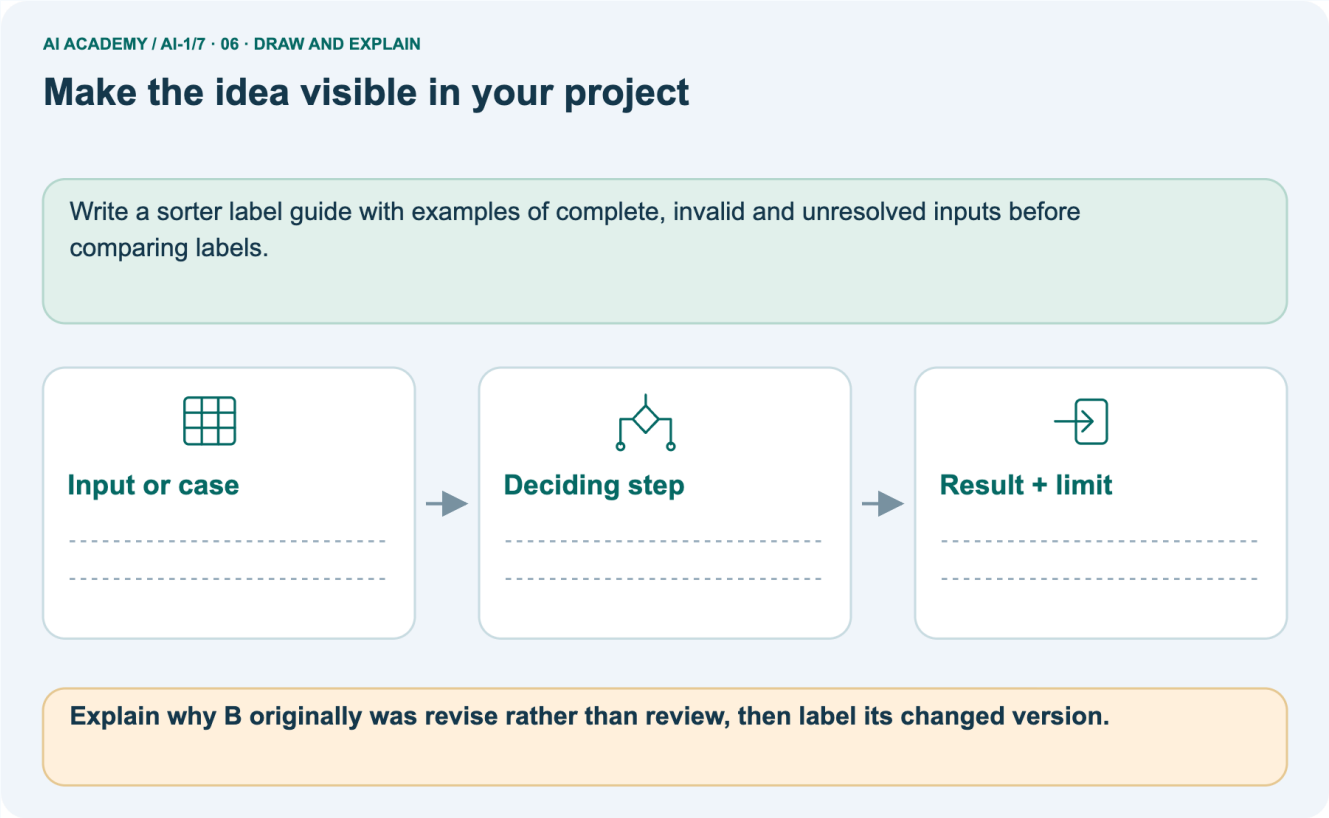
Operational **label**s specify what a word means for this task. Two **label**lers using the same priority should agree on these cards. Review describes missing **evidence**, not a hidden assumption that the card is wrong.

Revise is supported by a known failure. Limit: Agreement between **label**lers does not prove this policy is suitable for every school or that **label**s are objective truths.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Write a sorter **label** guide with examples of complete, invalid and unresolved **input**s before comparing **label**s. Draw your **input**, deciding step, result and limit.

Supported question: Explain why B originally was revise rather than review, then **label** its changed version.

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Categories and Labels: the case

02. Follow the deciding steps

03. Three ordered checks for a calendar card

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.